CCC-Code-without-Customer-ID.R

arppi

2026-08-06

```
#Uploaded Dataset
cc <- read.csv("E:/Customer Churn Capstone Project IIM/customer_churn_dataset-testing-master.cs
v", stringsAsFactors=TRUE)

#1 Data Preprocessing

summary(cc)
```

```
##    CustomerID         Age           Gender          Tenure      Usage.Frequency
##  Min.   :    1   Min.   :18.00   Female:34353   Min.   : 1.00   Min.   : 1.00
##  1st Qu.:16094   1st Qu.:30.00   Male  :30021   1st Qu.:18.00   1st Qu.: 7.00
##  Median :32188   Median :42.00                  Median :33.00   Median :15.00
##  Mean   :32188   Mean   :41.97                  Mean   :31.99   Mean   :15.08
##  3rd Qu.:48281   3rd Qu.:54.00                  3rd Qu.:47.00   3rd Qu.:23.00
##  Max.   :64374   Max.   :65.00                  Max.   :60.00   Max.   :30.00
##  Support.Calls    Payment.Delay   Subscription.Type  Contract.Length
##  Min.   : 0.000   Min.   : 0.00   Basic   :21451    Annual   :21410
##  1st Qu.: 3.000   1st Qu.:10.00   Premium :21421    Monthly  :22130
##  Median : 6.000   Median :19.00   Standard:21502    Quarterly:20834
##  Mean   : 5.401   Mean   :17.13
##  3rd Qu.: 8.000   3rd Qu.:25.00
##  Max.   :10.000   Max.   :30.00
##   Total.Spend   Last.Interaction     Churn
##  Min.   : 100   Min.   : 1.0     Min.   :0.0000
##  1st Qu.: 313   1st Qu.: 8.0     1st Qu.:0.0000
##  Median : 534   Median :15.0     Median :0.0000
##  Mean   : 541   Mean   :15.5     Mean   :0.4737
##  3rd Qu.: 768   3rd Qu.:23.0     3rd Qu.:1.0000
##  Max.   :1000   Max.   :30.0     Max.   :1.0000
```

```
str(cc)
```

```
## 'data.frame':    64374 obs. of  12 variables:
##  $ CustomerID       : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ Age              : int  22 41 47 35 53 30 47 54 36 65 ...
##  $ Gender           : Factor w/ 2 levels "Female","Male": 1 1 2 2 1 2 1 1 2 2 ...
##  $ Tenure           : int  25 28 27 9 58 41 37 36 20 8 ...
##  $ Usage.Frequency  : int  14 28 10 12 24 14 15 11 5 4 ...
##  $ Support.Calls    : int  4 7 2 5 9 10 9 0 10 2 ...
##  $ Payment.Delay    : int  27 13 29 17 2 10 28 18 8 23 ...
##  $ Subscription.Type: Factor w/ 3 levels "Basic","Premium",..: 1 3 2 2 3 2 1 3 1 1 ...
##  $ Contract.Length  : Factor w/ 3 levels "Annual","Monthly",..: 2 2 1 3 1 2 3 2 2 1 ...
##  $ Total.Spend      : int  598 584 757 232 533 500 574 323 687 995 ...
##  $ Last.Interaction : int  9 20 21 18 18 29 14 16 8 10 ...
##  $ Churn            : int  1 0 0 0 0 0 1 0 0 0 ...
```

```
#2 Lable Encoding and Dummy Variables

cc$Gender <-ifelse(cc$Gender=="Male",1,0)

library(fastDummies)
cc = dummy_columns(cc,select_columns = c("Subscription.Type","Contract.Length"))

#3 Correlation

#install.packages("pheatmap")

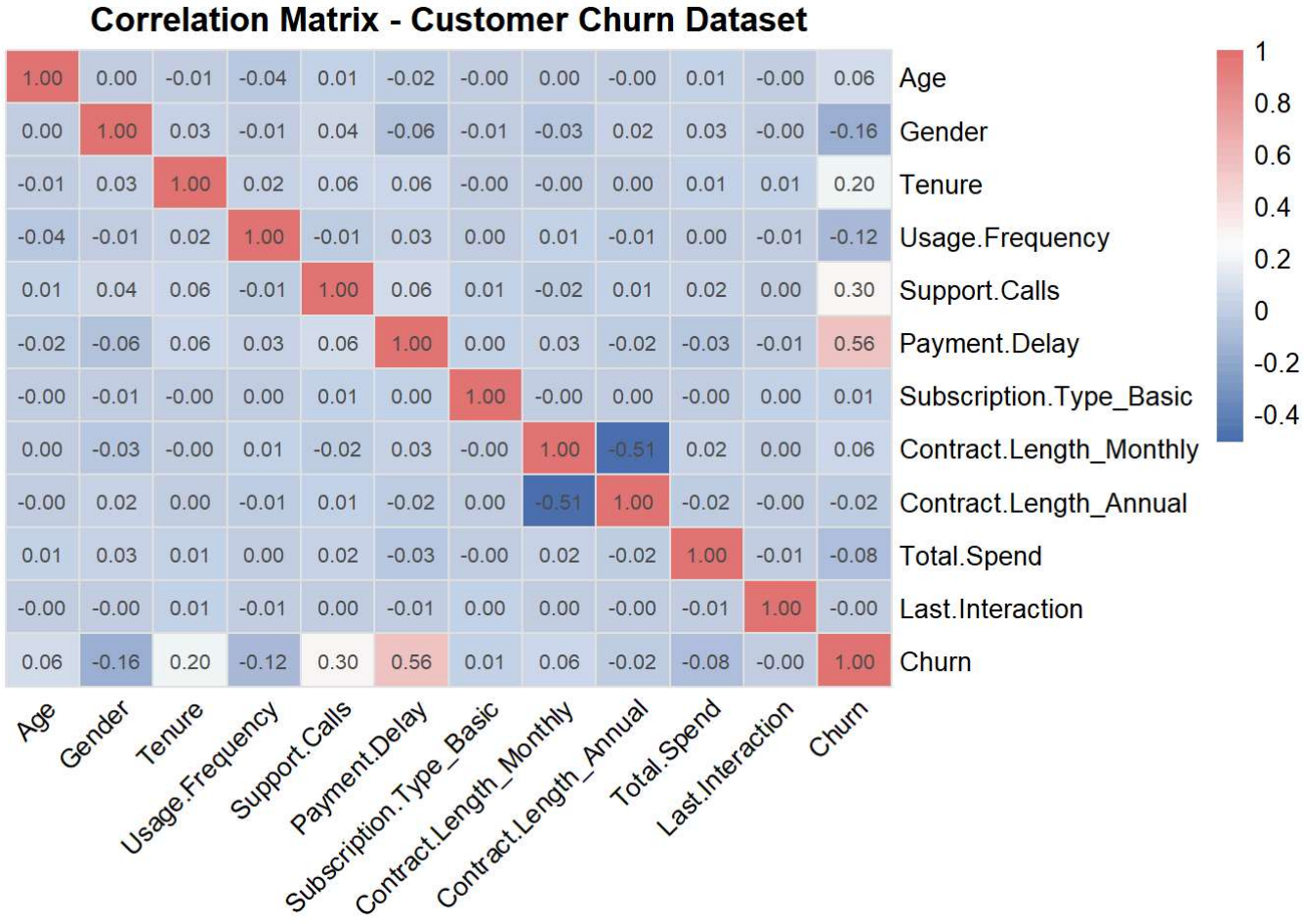
library(pheatmap)
```

```
## Warning: package 'pheatmap' was built under R version 4.5.3
```

```r
# Select variables
corr_data <- cc[, c(
  "Age",
  "Gender",
  "Tenure",
  "Usage.Frequency",
  "Support.Calls",
  "Payment.Delay",
  "Subscription.Type_Basic",
  "Contract.Length_Monthly",
  "Contract.Length_Annual",
  "Total.Spend",
  "Last.Interaction",
  "Churn"
)]

cr <- cor(corr_data)

pheatmap(
  cr,
  color = colorRampPalette(c("#4C72B0", "white", "#E57373"))(100),
  #breaks = seq(-1, 1, length.out = 101)),
  display_numbers = TRUE,
  number_format = "%.2f",
  cluster_rows = FALSE,
  cluster_cols = FALSE,
  border_color = "grey90",
  fontsize = 10,
  fontsize_number = 8,
  angle_col = 45,
  main = "Correlation Matrix - Customer Churn Dataset"
)
```




```
#Step 4: Spliting Dataset into Training and Testing Subsets
library(caTools)
set.seed(100)
split = sample.split(cc$Churn,SplitRatio = 0.8)
Trainingcc = subset(cc, split=='TRUE')
Testingcc = subset(cc,split=='FALSE')



#Step 5: Building Model on Training Data Subset

library(car)
```

```
## Loading required package: carData
```

```
#Forward Regression
model1 = glm(Churn ~ Payment.Delay, data = Trainingcc,  family= binomial)
summary(model1)
```

```
##
## Call:
## glm(formula = Churn ~ Payment.Delay, family = binomial, data = Trainingcc)
##
## Coefficients:
##                Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -3.084498   0.030314  -101.8   <2e-16 ***
## Payment.Delay  0.167170   0.001522   109.8   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 71250  on 51498  degrees of freedom
## Residual deviance: 53009  on 51497  degrees of freedom
## AIC: 53013
##
## Number of Fisher Scoring iterations: 4
```

```
#Residual deviance: 53009, AIC: 53013
model2 = glm(Churn ~ Payment.Delay + Support.Calls , data = Trainingcc, family=binomial)
summary(model2)
```

```
##
## Call:
## glm(formula = Churn ~ Payment.Delay + Support.Calls, family = binomial,
##     data = Trainingcc)
##
## Coefficients:
##                Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -4.994556   0.046034 -108.50   <2e-16 ***
## Payment.Delay  0.186321   0.001712  108.81   <2e-16 ***
## Support.Calls  0.284658   0.004033   70.58   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 71250  on 51498  degrees of freedom
## Residual deviance: 47160  on 51496  degrees of freedom
## AIC: 47166
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model2)
```

```
## Payment.Delay Support.Calls
##      1.132808      1.132808
```

```
#Residual deviance: 47160, AIC:47166

model3 = glm(Churn ~ Payment.Delay + Support.Calls+Tenure , data = Trainingcc, family=binomial)
summary(model3)
```

```
##
## Call:
## glm(formula = Churn ~ Payment.Delay + Support.Calls + Tenure,
##     family = binomial, data = Trainingcc)
##
## Coefficients:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -5.9939666  0.0550918 -108.80   <2e-16 ***
## Payment.Delay  0.1903991  0.0017609  108.12   <2e-16 ***
## Support.Calls  0.2839334  0.0041121   69.05   <2e-16 ***
## Tenure         0.0289345  0.0007019   41.22   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 71250  on 51498  degrees of freedom
## Residual deviance: 45374  on 51495  degrees of freedom
## AIC: 45382
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model3)
```

```
## Payment.Delay Support.Calls        Tenure
##      1.155118      1.127178      1.029233
```

```
#Residual deviance: 45374, AIC: 45382

model4 = glm(Churn ~ Payment.Delay + Support.Calls + Tenure+ Age , data = Trainingcc, family=bin
omial)
summary(model4)
```

```
##
## Call:
## glm(formula = Churn ~ Payment.Delay + Support.Calls + Tenure +
##     Age, family = binomial, data = Trainingcc)
##
## Coefficients:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -6.8956365  0.0703867  -97.97   <2e-16 ***
## Payment.Delay  0.1936383  0.0017887  108.26   <2e-16 ***
## Support.Calls  0.2876326  0.0041505   69.30   <2e-16 ***
## Tenure         0.0294211  0.0007071   41.61   <2e-16 ***
## Age            0.0191886  0.0008515   22.54   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 71250  on 51498  degrees of freedom
## Residual deviance: 44857  on 51494  degrees of freedom
## AIC: 44867
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model4)
```

```
## Payment.Delay Support.Calls        Tenure           Age
##      1.177544      1.133248      1.032058      1.022325
```

```
#Residual deviance: 44857, AIC: 44867

model5 = glm(Churn ~ Payment.Delay + Support.Calls + Tenure + Age+Contract.Length_Monthly , data
= Trainingcc, family=binomial)
summary(model5)
```

```
##
## Call:
## glm(formula = Churn ~ Payment.Delay + Support.Calls + Tenure +
##     Age + Contract.Length_Monthly, family = binomial, data = Trainingcc)
##
## Coefficients:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)              -7.0670187  0.0719622  -98.20   <2e-16 ***
## Payment.Delay             0.1941614  0.0017959  108.11   <2e-16 ***
## Support.Calls             0.2906507  0.0041775   69.58   <2e-16 ***
## Tenure                    0.0297208  0.0007102   41.85   <2e-16 ***
## Age                       0.0192908  0.0008542   22.58   <2e-16 ***
## Contract.Length_Monthly   0.3788403  0.0248769   15.23   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 71250  on 51498  degrees of freedom
## Residual deviance: 44623  on 51493  degrees of freedom
## AIC: 44635
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model5)
```

```
##           Payment.Delay           Support.Calls                  Tenure
##                1.182291                1.139733                1.033915
##                     Age Contract.Length_Monthly
##                1.022661                1.008041
```

```
#Residual deviance: 44623, AIC: 44635

model6 = glm(Churn ~ Payment.Delay + Support.Calls +
               Tenure + Age + Contract.Length_Monthly+ Subscription.Type_Basic, data = Trainingc
c, family=binomial)
summary(model6)
```

```
##
## Call:
## glm(formula = Churn ~ Payment.Delay + Support.Calls + Tenure +
##     Age + Contract.Length_Monthly + Subscription.Type_Basic,
##     family = binomial, data = Trainingcc)
##
## Coefficients:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)              -7.0923201  0.0725478 -97.761  < 2e-16 ***
## Payment.Delay             0.1941795  0.0017963 108.102  < 2e-16 ***
## Support.Calls             0.2906515  0.0041781  69.566  < 2e-16 ***
## Tenure                    0.0297261  0.0007102  41.854  < 2e-16 ***
## Age                       0.0192976  0.0008543  22.590  < 2e-16 ***
## Contract.Length_Monthly   0.3788146  0.0248783  15.227  < 2e-16 ***
## Subscription.Type_Basic   0.0732227  0.0249686   2.933  0.00336 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 71250  on 51498  degrees of freedom
## Residual deviance: 44615  on 51492  degrees of freedom
## AIC: 44629
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model6)
```

```
##           Payment.Delay           Support.Calls                  Tenure
##                1.182461                1.139832                1.033928
##                     Age Contract.Length_Monthly Subscription.Type_Basic
##                1.022663                1.008041                1.000155
```

```
#Residual deviance: 44615, AIC: 44629
#observation: with customer id Subscription.Type_Basic had 3 significant stars, but without it's
2 (still significant)
#Customer id Removed cause it's an identifier not a behavioral predictor/feature/Variable

model7 = glm(Churn ~ Payment.Delay + Support.Calls +
               Tenure + Age + Contract.Length_Monthly+ Subscription.Type_Basic+ Contract.Length_
Annual , data = Trainingcc, family=binomial)
summary(model7)
```

```
##
## Call:
## glm(formula = Churn ~ Payment.Delay + Support.Calls + Tenure +
##     Age + Contract.Length_Monthly + Subscription.Type_Basic +
##     Contract.Length_Annual, family = binomial, data = Trainingcc)
##
## Coefficients:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)              -7.1706576  0.0744881 -96.266  < 2e-16 ***
## Payment.Delay             0.1943410  0.0017975 108.115  < 2e-16 ***
## Support.Calls             0.2907553  0.0041796  69.565  < 2e-16 ***
## Tenure                    0.0297610  0.0007106  41.884  < 2e-16 ***
## Age                       0.0193132  0.0008546  22.600  < 2e-16 ***
## Contract.Length_Monthly   0.4520006  0.0289994  15.587  < 2e-16 ***
## Subscription.Type_Basic   0.0729166  0.0249748   2.920   0.0035 **
## Contract.Length_Annual    0.1434854  0.0291410   4.924 8.49e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 71250  on 51498  degrees of freedom
## Residual deviance: 44591  on 51491  degrees of freedom
## AIC: 44607
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model7)
```

```
##           Payment.Delay           Support.Calls                  Tenure
##                1.183529                1.139858                1.034250
##                     Age Contract.Length_Monthly Subscription.Type_Basic
##                1.022709                1.368877                1.000160
##  Contract.Length_Annual
##                1.359005
```

```
#Residual deviance: 44591, AIC: 44607

# Subscription.Type_Standard      not significant, increased AIC by , removed
# Last.Interaction                not significant, increased AIC by , removed
# Subscription.Type_Premium       not significant, increased AIC by , removed, also affecting th
e significance of Subscription.Type_Basic


model8 = glm(Churn ~ Payment.Delay + Support.Calls +
               Tenure + Age + Contract.Length_Monthly+ Subscription.Type_Basic+ Contract.Length_
Annual+ Total.Spend , data = Trainingcc, family=binomial)
summary(model8)
```

```
##
## Call:
## glm(formula = Churn ~ Payment.Delay + Support.Calls + Tenure +
##     Age + Contract.Length_Monthly + Subscription.Type_Basic +
##     Contract.Length_Annual + Total.Spend, family = binomial,
##     data = Trainingcc)
##
## Coefficients:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)              -6.732e+00  7.664e-02 -87.838  < 2e-16 ***
## Payment.Delay             1.959e-01  1.815e-03 107.912  < 2e-16 ***
## Support.Calls             2.969e-01  4.231e-03  70.171  < 2e-16 ***
## Tenure                    3.036e-02  7.167e-04  42.356  < 2e-16 ***
## Age                       1.959e-02  8.600e-04  22.782  < 2e-16 ***
## Contract.Length_Monthly   4.744e-01  2.921e-02  16.240  < 2e-16 ***
## Subscription.Type_Basic   7.307e-02  2.512e-02   2.908  0.00363 **
## Contract.Length_Annual    1.408e-01  2.933e-02   4.799 1.59e-06 ***
## Total.Spend              -1.003e-03  4.589e-05 -21.866  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 71250  on 51498  degrees of freedom
## Residual deviance: 44105  on 51490  degrees of freedom
## AIC: 44123
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model8)
```

```
##           Payment.Delay           Support.Calls                  Tenure
##                1.195611                1.152579                1.038270
##                     Age Contract.Length_Monthly Subscription.Type_Basic
##                1.023168                1.373758                1.000179
##  Contract.Length_Annual             Total.Spend
##                1.360818                1.019426
```

```
#Residual deviance: 44105, AIC: 44123


model9 = glm(Churn ~ Payment.Delay + Support.Calls +
               Tenure + Age + Contract.Length_Monthly+ Subscription.Type_Basic
             + Contract.Length_Annual+ Total.Spend+ Usage.Frequency , data = Trainingcc, family=
binomial)
summary(model9)
```

```
##
## Call:
## glm(formula = Churn ~ Payment.Delay + Support.Calls + Tenure +
##     Age + Contract.Length_Monthly + Subscription.Type_Basic +
##     Contract.Length_Annual + Total.Spend + Usage.Frequency, family = binomial,
##     data = Trainingcc)
##
## Coefficients:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)              -6.166e+00  7.846e-02 -78.587  < 2e-16 ***
## Payment.Delay             2.073e-01  1.917e-03 108.120  < 2e-16 ***
## Support.Calls             3.082e-01  4.362e-03  70.655  < 2e-16 ***
## Tenure                    3.245e-02  7.374e-04  43.999  < 2e-16 ***
## Age                       1.898e-02  8.778e-04  21.624  < 2e-16 ***
## Contract.Length_Monthly   4.969e-01  2.990e-02  16.618  < 2e-16 ***
## Subscription.Type_Basic   8.061e-02  2.568e-02   3.139  0.00169 **
## Contract.Length_Annual    1.359e-01  2.995e-02   4.537 5.71e-06 ***
## Total.Spend              -1.047e-03  4.697e-05 -22.290  < 2e-16 ***
## Usage.Frequency          -5.735e-02  1.430e-03 -40.103  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 71250  on 51498  degrees of freedom
## Residual deviance: 42398  on 51489  degrees of freedom
## AIC: 42418
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model9)
```

```
##           Payment.Delay           Support.Calls                  Tenure
##                1.278819                1.171409                1.051353
##                     Age Contract.Length_Monthly Subscription.Type_Basic
##                1.022630                1.375588                1.000276
##  Contract.Length_Annual             Total.Spend         Usage.Frequency
##                1.361114                1.021383                1.081508
```

```
#Residual deviance: 42398, AIC: 42418




model10 = glm(Churn ~ Payment.Delay + Support.Calls +
                Tenure + Age + Contract.Length_Monthly+ Subscription.Type_Basic
              + Contract.Length_Annual+ Total.Spend+ Usage.Frequency+ Gender , data = Trainingc
c, family=binomial)
summary(model10)
```

```
##
## Call:
## glm(formula = Churn ~ Payment.Delay + Support.Calls + Tenure +
##     Age + Contract.Length_Monthly + Subscription.Type_Basic +
##     Contract.Length_Annual + Total.Spend + Usage.Frequency +
##     Gender, family = binomial, data = Trainingcc)
##
## Coefficients:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)              -6.015e+00  8.071e-02 -74.526  < 2e-16 ***
## Payment.Delay             2.142e-01  1.998e-03 107.247  < 2e-16 ***
## Support.Calls             3.339e-01  4.596e-03  72.662  < 2e-16 ***
## Tenure                    3.522e-02  7.660e-04  45.981  < 2e-16 ***
## Age                       2.015e-02  9.046e-04  22.269  < 2e-16 ***
## Contract.Length_Monthly   4.837e-01  3.076e-02  15.726  < 2e-16 ***
## Subscription.Type_Basic   8.054e-02  2.640e-02   3.051  0.00228 **
## Contract.Length_Annual    1.558e-01  3.077e-02   5.064 4.11e-07 ***
## Total.Spend              -1.020e-03  4.824e-05 -21.138  < 2e-16 ***
## Usage.Frequency          -6.073e-02  1.475e-03 -41.165  < 2e-16 ***
## Gender                   -1.141e+00  2.597e-02 -43.939  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 71250  on 51498  degrees of freedom
## Residual deviance: 40342  on 51488  degrees of freedom
## AIC: 40364
##
## Number of Fisher Scoring iterations: 5
```

```
vif(model10)
```

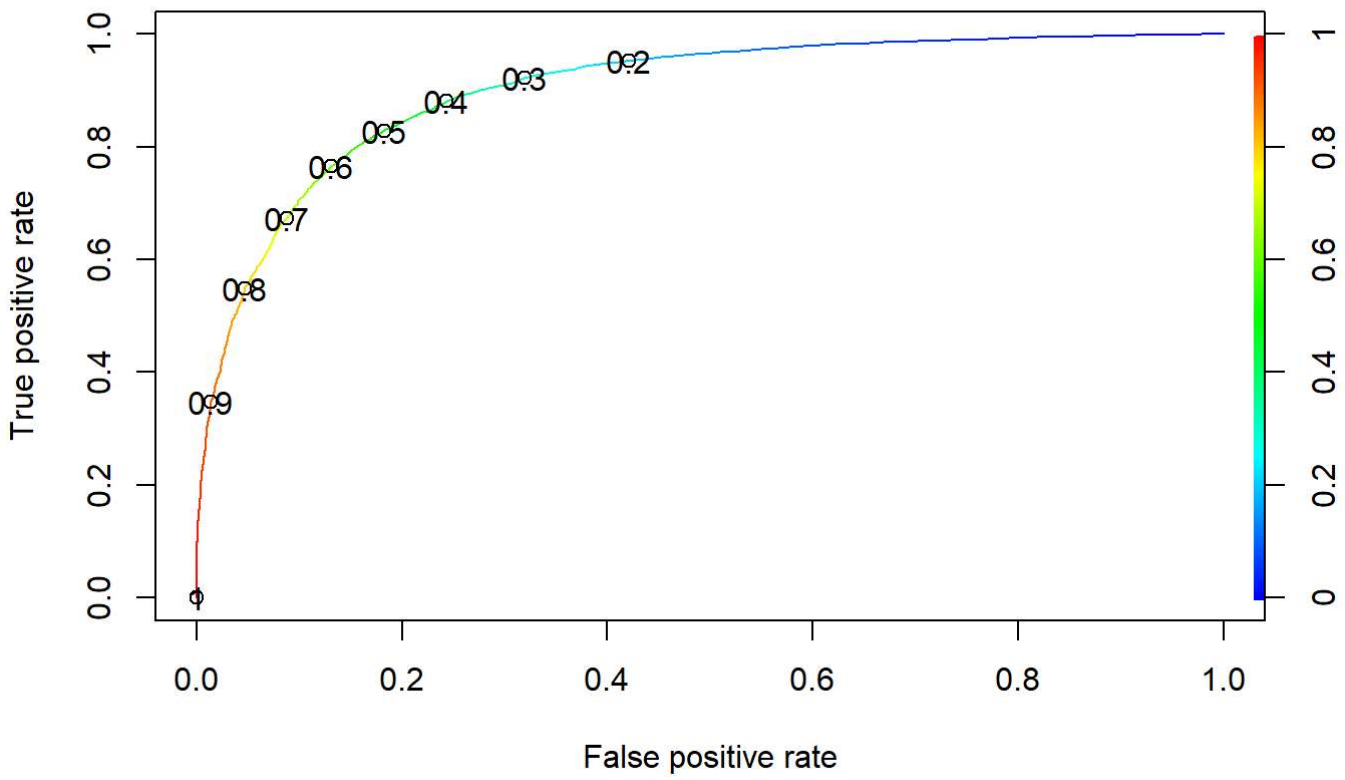
```
##           Payment.Delay           Support.Calls                  Tenure
##                1.332888                1.223957                1.069572
##                     Age Contract.Length_Monthly Subscription.Type_Basic
##                1.026109                1.375096                1.000261
##  Contract.Length_Annual             Total.Spend         Usage.Frequency
##                1.362280                1.020050                1.090912
##                  Gender
##                1.081256
```

```
#Residual deviance: 40342, AIC: 40364

#although Customer ID is only an identifier rather than behavioral feature, Even though it decre
ased the residual deviance and AIC enormously, one should not add it.

#Conclusion: We will choose model10 as our best model


#Step 6: ROCR Curve to decide the cutoff
library(ROCR)
Res = predict(model10,Testingcc,type = 'response')
#type = response will convert Log-R from Log into Probability
ROCRpred = prediction(Res,Testingcc$Churn)
#prediction function will calculate TP TN FP FN (aka Confusion Matrix outputs)
ROCRpref = performance(ROCRpred,"tpr","fpr")
#performance will calculate TPR FPR which you stored RPOCRpred
plot(ROCRpref,colorize=TRUE,print.cutoffs.at=seq(0.2,by=0.1))
```



```
#plot where colorise=TRUE, cuttoff will give probablity of TPR 0.8 &  FPR 0.3
# seq 0.2 means start from 0.2 and by0.1 means how much gap you want

#Step 7:Testing Data Subset on Testing Data, with Confusion Matrix


head(Res)
```

```
##          5          8          9         12         13         21
## 0.21149343 0.42147672 0.24722915 0.08254718 0.09082898 0.22184662
```

```
head(Testingcc$Churn)
```

```
## [1] 0 0 0 0 0 0
```

```
table(Actual=Testingcc$Churn,Predicted=Res>0.60)
```

```
##       Predicted
## Actual FALSE TRUE
##      0  5888  888
##      1  1427 4672
```

```
# builds a confusion matrix table with a treshold
mean(Testingcc$Churn == (Res > 0.50))
```

```
## [1] 0.8226019
```